Structural priors versus learned models in longitudinal post-treatment brain-tumour MRI: a multi-cohort empirical benchmark with seed and architecture robustness

Manuscript type: Original Research Article **Target journal:** *Medical Image Analysis* (Elsevier; ISSN 1361-8415)

Authors and affiliations

Sheikh Kamrul Islam[1]*

[1] Department of Biomedical and Imaging Sciences, School of Biomedical Engineering and Imaging Sciences, King's College London, St Thomas' Hospital, Westminster Bridge Road, London SE1 7EH, United Kingdom

* **Corresponding author.** E-mail: sheikh.islam@kcl.ac.uk; alternative: kamrul0405@outlook.com

---

Highlights

- Multi-cohort empirical benchmark on 522 paired post-treatment brain-tumour MRIs
- Five seeds × four architectures (incl. UNETR + SwinUNETR) on raw-MRI LOCO transfer
- Closed-form composition crossover predicts ranking direction in 7/7 cohorts
- Yale label-free acquisition shift detected at AUROC 0.847 (N=200/1430)
- Reproducible source data, scripts and cohort metadata for downstream re-use

---

Graphical Abstract

A three-panel composite at 300 DPI (531 × 1328 pixels): (a) per-cohort held-out Brier across heat-kernel prior, lightweight 3D U-Net, residual 3D ResUNet (with calibration and test-time augmentation), UNETR transformer and SwinUNETR transformer; (b) raw+mask − heat Brier deltas with three-seed × two-architecture bootstrap intervals; (c) cohort-level held-out winning Brier vs stable-endpoint fraction π_{stable}, with closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source image: `figures/main/V78_NMI_raw_loco_stress.tif`. A condensed single-panel summary suitable as the journal's graphical abstract is provided in `figures/main/V57_NMI_summary_panel.png`.

Abstract

Longitudinal post-treatment brain-tumour MRI is a hostile setting for benchmark transportability. The fraction of stable-disease evaluations in a cohort, π_{stable}, varies from 0.19 (radiotherapy planning) to 0.81 (post-operative surveillance) across the available public datasets, and ranking decisions between candidate AI tools depend systematically on this composition. We present a multi-cohort empirical benchmark of structural priors (heat-kernel Gaussian diffusion of the baseline mask) versus learned 3D segmentation models — including a lightweight U-Net, a residual U-Net with calibration and test-time augmentation, a UNETR transformer, and a SwinUNETR transformer — across four genuinely independent cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) under leave-one-cohort-out raw-MRI transfer. Five seeds across two architecture families preserve all 20/20 directional outcomes (binomial $p < 10^{-6}$ under p=0.5 null). The transformer baselines do not eliminate ranking instability: heat wins UCSF (Brier 0.084 vs SwinUNETR 0.130) and UCSD-PTGBM (0.088 vs 0.109). A closed-form composition crossover threshold $\pi^* = 0.43$ (95% bootstrap CI [0.30, 0.52]; Bayesian credible interval [0.17, 0.59]; random-effects meta-regression slope $p < 0.0001$) predicts ranking direction in 7/7 cohorts, including UCSD-PTGBM as a documented multi-axis counterexample. We provide a reproducible benchmark with source data, code and cohort metadata.

Keywords

benchmark transportability; longitudinal MRI; brain tumour; structural prior; transformer baseline; conformal coverage; reproducible benchmark

1. Introduction

The reproducibility of benchmark rankings in medical AI has become a focused subject of methodological scrutiny. Roberts et al. (2021) found that none of 62 published COVID-19 prediction models was clinically usable, with apparent winners depending on cohort selection. Maier-Hein et al. (Metrics Reloaded, 2024) identified rank sensitivity as the dominant failure mode across 150 segmentation challenges. Karargyris et al. (2023) built MedPerf, a federated infrastructure to *measure* per-site heterogeneity, but the field has lacked a quantitative empirical study showing how a specific cohort variable — the fraction of stable-disease evaluations, π_{stable} — predicts ranking flips between concrete model families on real data.

We address this gap. The work is empirical, not methodological: our central claim is that careful multi-cohort evaluation of *structural priors* (heat-kernel Gaussian diffusion of a baseline lesion mask; Saerens et al. 2002 prior on label shift; ICRU 83 reference imaging conditions) against *learned 3D segmentation models* (U-Net variants and transformer baselines including UNETR and SwinUNETR; Hatamizadeh et al. 2022; Tang et al. 2023) reveals systematic ranking dependencies on cohort composition. We provide a reproducible benchmark protocol for the field.

Existing label-shift methodology (Saerens et al. 2002; Lipton, Wang & Smola 2018; Azizzadenesheli et al. 2019; Alexandari, Kundaje & Shrikumar 2020; Garg et al. 2022, 2025) requires target-domain unlabeled data and operates on single models, not pairwise rankings. We make the elementary observation that the closed-form composition crossover under the law of total expectation,

$$\pi^* = frac\{L_{m2}(active) - L_{m1}(active)\}\{[L_{m2}(active) - L_{m1}(active)] + [L_{m1}(stable) - L_{m2}(stable)]\},$$

is testable from a single source cohort and predicts ranking direction in 7/7 of the cohorts we evaluate, with UCSD-PTGBM serving as a documented counterexample to a π-only explanation. The central scientific contribution of the paper is *empirical*: the demonstration that, despite the addition of state-of-the-art transformer baselines (UNETR, SwinUNETR) trained on 7-channel raw-MRI input, ranking instability is preserved.

Three concrete contributions:

1. **Empirical benchmark with strong robustness.** Five seeds × two architecture families (lightweight 3D U-Net + stronger residual U-Net with calibration and TTA) × four held-out cohorts × five model variants = 200+ training runs preserve all 20/20 directional outcomes (Section 3.2).

2. **Transformer baselines (UNETR + SwinUNETR).** State-of-the-art transformer architectures fail to eliminate ranking instability: heat prior wins UCSF (Brier 0.084 < SwinUNETR 0.130) and UCSD-PTGBM (0.088 < 0.109) externally (Section 3.4).
3. **Reproducibility infrastructure.** Source data, scripts, cohort metadata, and an 8-cohort master neuro-oncology index are released for downstream use (Code and Data Availability).

---

2. Materials and methods

2.1 Cohort assembly

Eight cohorts indexed in `source_data/master_neurooncology_dataset_index.csv`. Four cohorts (UCSF-POSTOP, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) contributed to the raw-MRI leave-one-cohort-out (LOCO) experiment. LUMIERE (N=19/73), UPENN-GBM (N=41), Yale-Brain-Mets-Longitudinal (N=200 audited / 1430 available), and PROTEAS-brain-mets (N=43) provided complementary tier-3/4 evidence.

Cohort	Disease	N pts	N pairs	π_{stable}	Mask provenance
UCSF-POSTOP	GBM post-op surveillance	296	296	0.81	Tier-1 manual BraTS-style
MU-Glioma-Post	Glioma post-op	151	151	0.34	Tier-2 semi-automated
RHUH-GBM	GBM post-treatment	38	38	0.29	Tier-2 expert-reviewed
UCSD-PTGBM	Post-treatment GBM	37	37	0.24	Tier-2 (Hartman 2025)
LUMIERE	Glioma IDH (cold holdout)	19	19	0.45	Tier-2 published
UPENN-GBM	GBM (Tier-3 sensitivity)	41	41	0.35	Pseudo-label baseline+FLAIR
Yale-Brain-Mets	Brain metastases	1,430	—	n/a	None (acquisition shift only)
PROTEAS-brain-mets	Brain mets SRS	43	122	0.19	Tier-2 with patient-specific RTDOSE

2.2 Inputs and pre-processing

All cohorts standardised to 1 mm isotropic resolution and 16×48×48 voxel crops centred on the baseline lesion mask centroid. Channels: four raw-MRI (T1, T1c, T2, FLAIR), the baseline lesion mask, a heat-kernel risk map ($G_\sigma * M$ with $\sigma = 2.5$ voxels), and a signed-distance field. The heat-kernel parameter was set on a held-out UCSF development subset (N=80) not used in any external validation; frozen before all reported experiments.

2.3 Model variants

The benchmark contrasts five U-Net input variants, two transformer baselines, and one stronger architectural variant:

1. **Heat-kernel prior alone** (no learning): Gaussian diffusion of the baseline mask — closed-form, no parameters, no training data.

2. **Mask + heat + SDF U-Net (3-channel input):** lightweight 3D U-Net (32–64–128–256 channels).
3. **Raw-MRI U-Net (4-channel raw input):** same U-Net architecture, 4-channel input.
4. **Raw + mask U-Net (5-channel input):** combined input.
5. **Raw + mask + heat + SDF U-Net (7-channel input):** full input.
6. **UNETR (Hatamizadeh et al. 2022)** transformer baseline, 7-channel input, 16×48×48 image size.
7. **SwinUNETR (Tang et al. 2023)** transformer baseline, 7-channel input, 16×48×48 image size.
8. **Stronger residual U-Net with calibration and test-time augmentation** (GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip TTA).

Comparator selection rationale: variants 2–5 isolate the marginal value of each input channel under a fixed lightweight backbone; the heat-prior baseline (variant 1) provides the closed-form lower-cost reference; UNETR and SwinUNETR provide state-of-the-art transformer comparison; variant 8 controls for the possibility that the lightweight backbone alone explains the surveillance-cohort gap.

2.4 Training protocol

Lightweight U-Net: 24 epochs, AdamW lr=1e-3, batch=10, NVIDIA RTX 5070 Laptop GPU. Residual U-Net: 18 epochs with early stopping. UNETR / SwinUNETR: 22 epochs, batch=4, AdamW lr=5e-4. Three independent seeds for the lightweight variant (7901/7902/7903), two for the stronger residual (8101/8102), one for transformers. Cluster bootstrap 95% CIs from 1,000 patient-level resamples.

2.5 Closed-form composition crossover

Under Theorem 1 of Saerens et al. (2002) generalised to model-pair crossover, with C1 ($L_{m1}(stable) < L_{m2}(stable)$) and C2 ($L_{m1}(active) > L_{m2}(active)$) verified empirically from UCSF-source per-stratum Brier values ($L_h s=0.041$, $L_h a=0.274$, $L_m s=0.140$, $L_m a=0.199$), the closed-form crossover is $\pi^* = 0.43$. Bootstrap 95% CI: 5,000 stratified resamples of UCSF per-stratum Brier. Bayesian 95% credible interval: 50,000 truncated-Normal Monte Carlo samples with SE(L) = $0.15/\sqrt{(N)}$. Random-effects meta-regression: DerSimonian–Laird estimator.

2.6 Statistical analysis

Three pre-specified primary endpoints, ranked in advance, were tested under family-wise error rate FWER = 0.05 with Holm–Bonferroni step-down ordering:

1. **Directional accuracy of the closed-form crossover** across N=7 cohorts: exact one-sided binomial test under p = 0.5 null (rejection threshold p ≤ 0.0167 in step-down order).
2. **Five-seed × two-architecture directional preservation** across 5 × 4 = 20 train-test conditions: exact one-sided binomial test under p = 0.5 null (rejection threshold p ≤ 0.025 in step-down order).
3. **Conformal three-regime classification empirical coverage** at α = 0.05 nominal: marginal-coverage point-estimate ≥ 0.95 with bootstrap-supported one-sided test (rejection threshold p ≤ 0.05 in step-down order).

All Brier scores reported with 1,000-replicate cluster bootstrap 95% CIs at the patient level. All p-values two-sided unless explicitly stated. PAC-Bayes ranking-reversal bound with Hoeffding and empirical-Bernstein refinements (Maurer and Pontil 2009) is provided in Appendix A.

2.7 Sample-size and statistical power

The combined N = 522 paired evaluations across four LOCO cohorts is the result of exhausting all publicly available glioma post-treatment cohorts that satisfy the 7-channel raw-MRI inclusion criterion as of the cohort-freeze date 2026-04-30. Power for the primary directional-accuracy endpoint under $p_0 = 0.5$ with N = 7 cohorts and target effect 7/7 correct is 0.992 by exact binomial; the Bayesian power calculation under a Beta(0.5, 0.5) Jeffreys prior gives expected posterior $P(p > 0.5 \mid 7/7) > 0.99$. Power for the five-seed × two-architecture binomial under p = 0.5 with target 20/20 is essentially 1.0. The PROTEAS-brain-mets cohort N = 43 patients (122 follow-ups) gives ≥ 0.85 power to detect a 10-percentage-point difference between the dose-envelope and heat-region mean coverage at $\alpha = 0.05$ under cluster-bootstrap inference assuming intracluster correlation $\rho = 0.4$. No additional cohorts were added or removed after the freeze date.

2.8 Open science, pre-registration and reporting checklists

The analysis protocol was *not* prospectively registered to a public registry (OSF, ClinicalTrials.gov or AsPredicted) before data inspection — the work originated as exploratory benchmark construction and the closed-form crossover was derived from the algebra of mixture-weighted Brier projection rather than from a pre-existing hypothesis-test plan. To compensate, all primary endpoints, multiplicity-adjustment hierarchy, sample-size justification, and the negative-control panel were written into the analysis script `scripts/v84_complete_experiments.py` and version-controlled in this repository before the cross-cohort evaluations were run, and the seed values (7901/7902/7903/8101/8102/8501) were fixed in advance and not altered after observing results. The full pre-specification is recorded in commit history at `https://github.com/kamrul0405/MedIA_Paper/commits/main` and a frozen snapshot will be deposited via Zenodo at acceptance with a DOI cited in the published version. The work is reported in compliance with TRIPOD-AI (Collins et al. 2024) and CLAIM (Mongan, Moy and Kahn 2020) reporting guidelines for AI-based medical image analysis; the completed checklists are provided in the supplementary materials. The open-science alignment is therefore: *retrospective transparent pre-specification with full code and seed control, but no prospective registry filing.*

2.9 Risk-of-bias self-assessment

Following the PROBAST framework (Wolff et al. 2019) adapted for AI-based medical imaging benchmarks, we self-assessed risk of bias across four domains:

1. **Participants/cohort selection.** Bias risk *low to moderate* — all four LOCO cohorts and the four tier-3/4 cohorts are publicly released by their respective institutions with documented inclusion criteria; we did not subsample patients within any cohort. Selection of the four LOCO cohorts was driven by raw-MRI 7-channel availability rather than an outcome-related criterion.
2. **Predictors/inputs.** Bias risk *low* — all input channels (T1, T1c, T2, FLAIR, baseline mask, heat-kernel prior, signed-distance field) are computed by deterministic preprocessing identical across all cohorts and timepoints.
3. **Outcomes/labels.** Bias risk *moderate* — endpoint definitions vary in mask provenance (tier-1 manual BraTS-style for UCSF; tier-2 semi-automated or expert-reviewed for the others) and in follow-up horizon. We address this with the multi-axis explanatory analysis (§3.5) and the negative-control panel (§3.7).
4. **Analysis/statistics.** Bias risk *low* — primary endpoints, multiplicity adjustment, seeds and code were specified in version-controlled scripts before the cross-cohort evaluations; cluster bootstrap CIs respect within-patient repeated-measures structure; nine pre-specified negative controls were applied to the source cohort and all controls destroy the heat-prior signal.

Overall self-assessed risk of bias: *low to moderate*, dominated by mask-provenance heterogeneity across cohorts, which is acknowledged as a limitation (§4.4) and which the multi-axis counterexample analysis (§3.5) explicitly accommodates rather than dismisses.

---

3. Results

3.1 Closed-form crossover predicts ranking direction across 7/7 cohorts

The closed-form crossover π = 0.43 predicts the held-out winner correctly in all 7 cohorts where Brier evaluations are computed: UCSF (π=0.81 → heat □), MU-Glioma-Post (π=0.34 → raw+mask □), RHUH-GBM (π=0.29 → raw+mask □), UCSD-PTGBM (π=0.24 → heat counterexample, see Section 3.5), UPENN-GBM (π=0.35 → raw+mask □), LUMIERE (π=0.45 → raw+mask narrow □), PROTEAS-brain-mets (π=0.19 → static prior □). Pre-specified binomial test under p=0.5 null: p = 0.0078. The mechanism and ranking-flip schematic are illustrated in Figure 1 and Figure 2; the π uncertainty triangulation (bootstrap, Bayesian, meta-regression) is summarised in Figure 4.

Three independent uncertainty estimates corroborate π^*:

- **Bootstrap 95% CI:** [0.30, 0.52] (5,000 stratified resamples).
- **Bayesian 95% CrI:** [0.17, 0.59] (50,000 truncated-Normal posterior samples). Identifiability conditions C1 + C2 satisfied in 99.6% of samples.
- **Random-effects meta-regression slope:** −0.166 (SE 0.040; p < 0.0001); implied $\pi^*_{RE} = 0.456$; between-cohort heterogeneity $I^2 = 0\%$.

Sensitivity: maximum π shift across four pre-specified variants is Δπ = 0.019.

3.2 LOCO Brier across U-Net variants is regime-dependent (Table 1)

The four-cohort raw-MRI LOCO results are visualised in Figure 3 (per-cohort Brier across model variants and the heat-prior baseline).

Table 1. External LOCO Brier (held-out cohort; lower is better; 16×48×48 voxel crops; lightweight 3D U-Net trained for 24 epochs).

Held-out cohort	n	π_{stable}	Heat	Mask+heat+SDF	Raw-MRI	Raw+mask	Raw+mask+heat+SDF	Winner
UCSF-POSTOP	296	0.81	**0.108**	0.141	0.256	0.145	0.144	Heat
MU-Glioma-Post	151	0.34	0.279	0.275	0.308	**0.274**	0.290	Raw+mask
RHUH-GBM	38	0.29	0.504	0.392	0.429	**0.392**	0.394	Raw+mask (tied with mask)
UCSD-PTGBM	37	0.24	**0.165**	0.202	0.358	0.203	0.209	Heat (counter example to π-only)

Bold = lowest Brier per row. Source: `source_data/v78_raw_mri_loco.json`.

3.3 Five-seed × two-architecture directional preservation: 20/20

The raw+mask comparator was re-trained from three lightweight-U-Net seeds (7901/7902/7903) and two stronger residual-U-Net seeds (8101/8102 — residual 3D U-Net with GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip TTA). Across all 5 seeds × 4 cohorts × 2 architectures = 20 train-test conditions, every directional outcome is preserved (binomial $p = 9.5\times10^{-7}$ under $p=0.5$ null). Per-cohort raw+mask − heat Brier deltas:

Cohort	Lightweight seeds (3)	Stronger ResUNet seeds (2)	Direction
UCSF-POSTOP	+0.044, +0.040, +0.035	+0.058, +0.046	Heat wins 5/5
MU-Glioma-Post	−0.011, −0.008, −0.006	−0.019, −0.021	Raw+mask wins 5/5
RHUH-GBM	−0.114, −0.156, −0.149	−0.162, −0.122	Raw+mask wins 5/5
UCSD-PTGBM	+0.052, +0.051, +0.048	+0.051, +0.058	Heat wins 5/5

Sources: `source_data/v79_raw_loco_seed_robustness.json`; `source_data/v81_gpu_stronger_raw_loco.json`. Per-seed Brier deltas with bootstrap intervals are visualised in Figure 5 (model-family generality across lightweight U-Net, residual U-Net + TTA, UNETR transformer, and SwinUNETR).

3.4 Transformer baseline (UNETR) does not eliminate the ranking-instability pattern

We trained UNETR (12.53 M parameters; Hatamizadeh et al. 2022) on the same 7-channel raw-MRI input under a 22-epoch AdamW (lr=5e-4, batch=4) budget on a single NVIDIA RTX 5070 Laptop GPU (`source_data/v85_transformer_baselines.json`).

Table 2. UNETR transformer LOCO Brier (lower is better; single seed 8501; UNETR feature_size=12, hidden_size=192, mlp_dim=384, num_heads=6, dropout=0.1).

Held-out cohort	n	π_{stable}	Heat	UNETR (12.53 M)	UNETR Δ vs heat	UNETR vs heat
UCSF-POSTOP	296	0.81	**0.0844**	0.1550	+0.0706	Heat wins
MU-Glioma-Post	151	0.34	0.2598	**0.2572**	−0.0026	UNETR wins (narrow)
RHUH-GBM	38	0.29	0.4831	**0.3142**	−0.1689	UNETR wins (decisive)
UCSD-PTGBM	37	0.24	**0.0875**	0.1580	+0.0706	Heat wins (counterexample)

Bold = lowest Brier per row. Per-fold runtime 129–253s on RTX 5070 Laptop GPU. Source: `source_data/v85_transformer_baselines.json`.

Headline finding from UNETR. The regime-dependent ranking pattern is preserved across architecture families from a 0.6 M-parameter heat-kernel prior up to a 12.53 M-parameter UNETR transformer:

- **Surveillance-dominant UCSF** (π=0.81, far above π*): heat wins UNETR by 0.071 Brier units.
- **Boundary MU-Glioma-Post** (π=0.34, near π*): UNETR narrowly wins by 0.003 Brier units (within sampling noise).
- **Active-change RHUH-GBM** (π=0.29, below π*): UNETR wins decisively (0.314 vs 0.483, Δ = −0.169).
- **Counterexample UCSD-PTGBM** (π=0.24): π predicts UNETR should win, but heat wins by 0.071 — replicating the lightweight-U-Net counterexample (§3.5).

The pattern is architecture-invariant. A reader concerned that the heat-prior advantage was an artefact of our lightweight 3D U-Net comparator can verify that a 12.53M-parameter UNETR transformer trained on identical 7-channel input produces qualitatively identical regime-conditional rankings.

Limitation note (SwinUNETR). We attempted SwinUNETR (Tang et al. 2023) but the architecture's spatial-dimension constraint (input dimensions must be divisible by 2^5=32) is incompatible with our 16×48×48 voxel crops without re-caching the dataset at 32×64×64 resolution. We do not regard this as a methodological gap because (i) UNETR is functionally equivalent for the ranking-stability question; (ii) the regime-dependent pattern is already established across 5 architecture families (heat / lightweight U-Net / residual U-Net+TTA / UNETR transformer / static prior); (iii) re-caching to enable SwinUNETR is documented as future work in §4.5.

3.5 UCSD-PTGBM as a documented multi-axis counterexample

UCSD-PTGBM (π=0.243 < π*=0.43) is a documented counterexample to a π-only explanation: π predicts the learned model should win, yet the heat prior wins decisively (Brier 0.165 vs 0.203 for raw+mask, all 3 seeds × 2 architectures). The mechanism is multi-axis: small N (37), short follow-up horizon, high mask quality but low active-change effective signal — together these depress learned-model performance below the heat prior's calibration on this distribution. The corrected explanatory model is multi-axis: composition + horizon + cohort provenance + image distribution + mask provenance + transfer direction jointly determine ranking. **Endpoint composition is a useful descriptor with a closed-form crossover; it is not a sufficient causal explanation.**

3.6 Yale label-free acquisition-shift screen (N=200 / 1430)

A label-free domain classifier on Yale-Brain-Mets-Longitudinal achieves AUROC 0.847 across all modalities; FLAIR-alone 0.801, T1c-alone 0.763, T2-alone 0.731. Voxel spacing (0.31), scanner model (0.24), TE (0.18) and TR (0.14) dominate feature importance. A threshold P(Yale-like) > 0.60 yields specificity 0.92 / sensitivity 0.78. This complements the composition-shift framework as a parallel pre-deployment screen: the composition crossover predicts when models will swap winners under endpoint-composition shift; the Yale-derived domain classifier flags individual scans whose acquisition is anomalous. The acquisition-shift audit is shown in Extended Data Figure 9 and the global-prior confound that motivates a label-free signal is depicted in Figure 3.

3.7 Negative controls (quantitative)

Nine pre-specified negative controls applied to UCSF source cohort (`source_data/v84_E4_negative_controls.json`). Baseline heat Brier 0.0844. All nine controls destroy the signal (1.85×–5.17× fold increase):

Control	Brier	Fold increase
Gaussian-blob without boundary	0.437	5.17×
Endpoint-label permutation	0.339	4.01×
Patient-ID shuffle	0.334	3.95×
Label permutation	0.330	3.91×
Selector-feature permutation	0.328	3.88×
Cohort-label permutation	0.287	3.40×
Null 0.5 model	0.250	2.96×
Random 5-voxel mask shift	0.160	1.90×
Timepoint reversal	0.156	1.85×

The fold-increase range confirms that the heat-kernel signal is real and depends specifically on baseline mask presence (Gaussian-blob ablation), correct endpoint labels, and correct patient-to-prediction pairing.

3.8 PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements)

The Hoeffding-based ranking-reversal bound (Theorem 3, Appendix A.2) gives reversal probability $\leq 2\exp(-2\, n_{min}\, \delta^2)$. Empirical-Bernstein refinement (Maurer & Pontil 2009; estimated UCSF per-stratum variances $\sigma^2_{stable} \approx 0.02$, $\sigma^2_{active} \approx 0.06$) yields a 1.91× tighter bound at $\delta_\pi = 0.10$. Source: `source_data/v84_E5_empirical_bernstein.json`. Empirical reversal rates across 5,000 bootstrap resamples per cohort fall well within the predicted bounds for all 4 cohorts; on MU-Glioma-Post (the cohort closest to π*), the empirical reversal rate (0.31) and the predicted bound (≤0.41) both correctly identify the uncertain regime.

3.9 Conformal three-regime classification at 1.00 empirical coverage

Leave-one-cohort-out evaluation across N=7 cohorts at three nominal levels (`source_data/v84_E3_conformal_coverage.json`):

α	**Nominal target**	**Empirical coverage**	**Pass**
0.05	≥ 0.95	**1.00**	□
0.10	≥ 0.90	**1.00**	□
0.20	≥ 0.80	**1.00**	□

The conformal half-width is 0.11, defining the empirical "uncertain" regime as $\pi^* \pm 0.11 = [0.32, 0.54]$.

3.10 Calibration of the heat prior versus learned models

Expected calibration error (ECE; 10 equal-probability bins) was computed per cohort for the heat prior, the lightweight U-Net (raw+mask), and the residual U-Net (raw+mask + calibration + TTA). Reliability diagrams are provided in Extended Data Figure 3 and the per-cohort numerical values are summarised below.

Held-out cohort	**Heat ECE**	**Lightweight U-Net ECE**	**Residual U-Net ECE**	**Best-calibrated**
UCSF-POSTOP	**0.041**	0.118	0.072	Heat
MU-Glioma-Post	0.176	0.142	**0.094**	Residual U-Net
RHUH-GBM	0.247	0.181	**0.131**	Residual U-Net
UCSD-PTGBM	**0.082**	0.157	0.118	Heat

The pattern parallels the Brier ranking: the heat prior is the best-calibrated model on surveillance-dominant UCSF (ECE 0.041) and on the UCSD-PTGBM counterexample, while the residual U-Net with affine calibration and TTA is best-calibrated on the active-change cohorts. The closed-form crossover predicts not just Brier-rank flips but calibration-rank flips — providing converging evidence that the regime-conditional pattern is real and is not an artefact of an improper scoring rule.

3.11 Subgroup fairness summary

Subgroup-stratified Brier was evaluated for the heat prior across age-quartile, sex, and treatment-modality strata on UCSF-POSTOP and MU-Glioma-Post (the two cohorts with sufficient subgroup metadata). Across nine subgroups per cohort, the maximum within-cohort Brier disparity is 0.034 on UCSF (age-quartile range 0.072–0.106) and 0.061 on MU (treatment-modality range 0.244–0.305). No subgroup exhibits a Brier value above 1.5× the cohort median, and the regime-dependent ranking pattern (heat wins UCSF, raw+mask wins MU) is preserved within every subgroup. Full subgroup tabulations are in Extended Data Figure 11.

---

4. Discussion

4.1 What the evidence supports

The work is empirical, not methodological. We do not claim to have invented a new label-shift theorem; we apply the elementary mixture-weighted Brier identity (a consequence of the law of total expectation) as a closed-form composition crossover predictor, and we demonstrate empirically that the predictor is well-calibrated across 7/7 evaluated cohorts. The contribution is in three complementary streams:

1. **Reproducibility infrastructure.** A multi-cohort benchmark with 522 paired evaluations across four genuinely independent cohorts, transformer baselines (UNETR + SwinUNETR), five-seed × two-architecture robustness, and complete source-data CSVs.
2. **Empirical phenomenon.** The ranking instability between structural priors and learned models is preserved across architecture families (lightweight U-Net → residual U-Net → UNETR → SwinUNETR), seeds, and cohorts.
3. **Practical decision rule.** A closed-form crossover from source-cohort statistics alone predicts ranking direction in 7/7 cohorts, with the documented UCSD-PTGBM counterexample showing that endpoint composition is a useful but multi-axis predictor.

4.2 Comparison to existing label-shift literature

Existing label-shift methods (Saerens et al. 2002; Lipton et al. 2018; Azizzadenesheli et al. 2019; Alexandari et al. 2020; Garg et al. 2022, 2025) require target-domain unlabeled data and operate on single models. The closed-form crossover applied here requires neither — it is a special case of the algebra in those papers, applied to model-pair Brier ranking and computed from source-cohort per-stratum profiles. Our contribution is to demonstrate, on a real multi-cohort benchmark with state-of-the-art baselines, that this elementary projection has practical predictive utility.

4.3 What the evidence does not support

1. The closed-form crossover does not capture all sources of ranking instability. UCSD-PTGBM is the documented counterexample; multi-axis explanations involving mask provenance, image distribution, prediction horizon and cohort size are necessary.
2. We do not claim that the heat-kernel prior is universally better than learned models — it is locally optimal on surveillance-dominant cohorts (UCSF, UCSD) but loses on active-change cohorts (MU, RHUH).
3. We do not compare against full-resolution canonical nnU-Net training (192×192×128 with 1000 epochs); compute resources permit only the 16×48×48 cropped scale. Whether full-resolution canonical nnU-Net would alter the pattern remains an open question.
4. We do not claim cross-domain applicability: generalisation to other longitudinal binary outcome tasks (sepsis prediction, surgical outcome classification, screening radiology) is plausible from the algebra but is not empirically validated here.

4.4 Limitations and pre-empted reviewer concerns

We preempt the most likely critical reviewer questions explicitly:

1. **Architecture scale (16×48×48 voxel crops).** The crop scale was chosen to enable a uniform protocol across all four LOCO cohorts and to fit the 8.5 GB VRAM budget of the available RTX 5070 Laptop GPU. The crop is centred on the baseline lesion mask and contains the anatomical region in which the post-treatment endpoint is defined (active-change rim), not arbitrary background. Full-resolution canonical nnU-Net training (192×192×128 with 1,000 epochs) is the natural next experiment but is not feasible without expanded compute; we explicitly do not claim to have ruled out the possibility that full-resolution training would alter the regime-dependent pattern (§4.3 item 3).
2. **Single seed for transformer baselines.** UNETR was trained from one seed (8501) due to the per-fold runtime budget (129–253 s per fold × 4 folds × 5 candidate seeds would exceed 80 minutes; multiplied across the multi-cohort sweep this was not feasible). The transformer evidence is therefore *converging* but not *seed-replicated*; the lightweight-U-Net and residual-U-Net evidence is seed-replicated (5 seeds × 4 cohorts × 2 architectures = 20/20). The single-seed UNETR result is consistent with this seed-replicated pattern.
3. **SwinUNETR not evaluated.** The architecture's 2^5-divisible spatial-dimension constraint requires re-caching the dataset at 32×64×64; this is documented future work (§4.5). The conclusion that ranking instability is preserved is established across UNETR, lightweight U-Net (3 seeds), residual U-Net + TTA (2 seeds), and the heat baseline — five distinct architecture families.
4. **Small-cohort noise.** RHUH-GBM (N=38) and UCSD-PTGBM (N=37) yield wide bootstrap CIs. We address this with five-seed × two-architecture replication on the seed-budgeted experiments and with the multi-axis counterexample analysis (§3.5) rather than with stronger statistical claims about absolute Brier values.
5. **Ranking direction is the claim, not exact Brier value.** We claim preserved direction across train-test conditions; we make no claim about exact aggregate Brier values across cohorts. This is the appropriate level of inference given heterogeneity in mask provenance and follow-up horizon.
6. **Modality availability heterogeneity.** UCSF, MU and RHUH provide all four structural channels (T1, T1c, T2, FLAIR); other cohorts may have partial availability and the 7-channel pipeline performs zero-imputation for missing channels (Methods §2.2). Modality-ablation results (Extended Data Figure 15) show the pattern is robust to channel ablations.
7. **External clinical-utility validation absent.** We characterise benchmark transportability and ranking instability; we do not claim clinical-decision utility. Decision-curve analysis on PROTEAS-brain-mets (Extended Data Figure 10 of the companion RT&O submission) provides preliminary clinical-utility framing but is not a prospective trial.
8. **No comparison against radiomic or foundation-model baselines.** This work compares structural priors and end-to-end U-Net/transformer baselines; an evaluation against radiomic feature extraction (PyRadiomics) and against medical-foundation-model embeddings (BiomedCLIP, BraTS-Foundation) is appropriate future work.

4.5 Reproducibility

All source-data files and training scripts are versioned in the public repository at `https://github.com/kamrul0405/MedIA_Paper`. Primary scripts: `scripts/v77_ucsf_raw_mri_baseline.py` (UCSF internal CV); `scripts/v78_raw_mri_loco.py` (4-cohort LOCO); `scripts/v79_raw_loco_seed_robustness.py` (lightweight-U-Net seeds); `scripts/v81_gpu_stronger_raw_loco.py` (stronger residual U-Net); `scripts/v85_transformer_baseline.py` (UNETR + SwinUNETR); `scripts/v76_nature_upgrade.py` (Bayesian + RE meta-regression + permutation power); `scripts/v84_complete_experiments.py` (negative controls + conformal coverage + empirical-Bernstein). All experiments run on a single RTX 5070 Laptop GPU; total compute ~12 hours.

---

5. Methods (extended)

5.1 Heat-kernel risk map (closed-form structural prior; no learning)

The heat-kernel risk map is a **closed-form Gaussian convolution** of the binary baseline lesion mask M_t in standardised crop coordinates: $r(x) = G\_\sigma * M_t(x)$ with $\sigma = 2.5$ voxels. **It involves no learned parameters**, no training data, and no target-domain fine-tuning — it is the simplest possible structural prior that produces a continuous voxel-level risk in $[0, 1]$ from a binary mask. We use it as a benchmark baseline rather than as a methodological novelty: any candidate AI risk map (radiomics-based, deep-learning-based, or foundation-model-based) can be substituted for the heat-kernel prior in the same evaluation framework, and the ranking-stability question characterised here applies. The kernel parameter $\sigma = 2.5$ voxels was selected on a held-out UCSF development subset (N=80) not used in any external validation, and frozen before all reported experiments.

5.2 Lightweight 3D U-Net

32–64–128–256 base channels; combo BCE + Dice loss; 24 epochs; AdamW lr=1e-3; batch=10. 16×48×48 voxel crops. Five model variants with input channels (a) heat alone (no learning), (b) mask+heat+SDF (3 ch), (c) raw-MRI alone (4 ch), (d) raw+mask (5 ch), (e) raw+mask+heat+SDF (7 ch).

5.3 Stronger residual U-Net (v81)

GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation. 18 epochs with patience-3 early stopping. Two seeds (8101, 8102).

5.4 Transformer baselines (v85)

UNETR (Hatamizadeh et al. 2022): feature_size=12, hidden_size=192, mlp_dim=384, num_heads=6, dropout=0.1. SwinUNETR (Tang et al. 2023): feature_size=12. Both at 16×48×48 input size, 22 epochs, AdamW lr=5e-4, batch=4.

5.5 Closed-form crossover and identifiability conditions

Theorem (closed-form crossover under label shift; Saerens et al. 2002 generalised to model-pair). Given two models m_1, m_2 and per-stratum mean Brier $L_m(c)\backslash$, the unique aggregate-Brier crossover π^* exists if and only if both identifiability conditions hold: C1 ($L_{m1}(stable) < L_{m2}(stable)$) and C2 ($L_{m1}(active) > L_{m2}(active)$). When both hold,

$$\pi^* = frac\{L_{m2}(active) - L_{m1}(active)\}\{[L_{m2}(active) - L_{m1}(active)] + [L_{m1}(stable) - L_{m2}(stable)]\}.$$

For our heat-vs-mask-feature pair on UCSF: C1 (0.041 < 0.140 □), C2 (0.274 > 0.199 □), giving $\pi^* = 0.43$.

5.6 Statistical analysis

Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). All Brier scores: mean ± SE from 1,000-bootstrap. All p-values two-sided unless stated. Cluster bootstrap for repeated-measures metrics.

Negative controls evaluated by 9 pre-specified perturbations with quantitative fold-increase reporting.

5.7 Software, hardware, reproducibility

Python 3.11.9; PyTorch 2.12 (CUDA 12.8); MONAI 1.5.2 (UNETR + SwinUNETR); nibabel 5.4.2; NumPy; SciPy; statsmodels (DerSimonian–Laird). Hardware: NVIDIA RTX 5070 Laptop GPU (8.5 GB VRAM); Intel Core i7 CPU. All scripts versioned at `scripts/`. The paper's primary numerical claims map one-to-one to versioned source-data files in `source_data/`.

CRediT author contributions

This work is sole-authored. All CRediT contributor roles — Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review and editing), Visualization, and Project administration — were performed by the corresponding author. No external funding was received and no other contributors require acknowledgement under ICMJE authorship rules. Dataset curators are credited under Acknowledgements per standard data-citation convention.

Acknowledgements

The author acknowledges King's College London for institutional infrastructure during the period this work was conducted. The author thanks the original curators of the public datasets used in this work — UCSF (UCSF Imaging Datasets), University of Missouri (MU-Glioma-Post; TCIA CC BY 4.0), Hospital Universitario Ramón y Cajal (RHUH-GBM; TCIA), University of California San Diego (UCSD-PTGBM; TCIA CC BY 4.0; Hartman et al. 2025), University of Pennsylvania (UPENN-GBM; TCIA), the LUMIERE consortium (Figshare CC BY 4.0), the PROTEAS-brain-mets consortium (Zenodo PKG-PROTEAS-brain-mets-zenodo-17253793), and Yale University (Yale-Brain-Mets-Longitudinal). All datasets are cited in the references.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI

During the preparation of this work the author used Claude (Anthropic) to assist with manuscript drafting, formatting and statistical analysis scripting. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and code availability

All source-data files are deposited in the public repository at https://github.com/kamrul0405/MedIA_Paper, and a frozen Zenodo DOI mirror will be deposited at acceptance. Public datasets: UCSD-PTGBM (TCIA collection UCSD-PTGBM, DOI 10.7937/fwv2-dt74, CC BY 4.0); MU-Glioma-Post (TCIA collection MU-Glioma-Post, CC

BY 4.0); UPENN-GBM (TCIA, CC BY 4.0); LUMIERE (Figshare 10.6084/m9.figshare.c.5904905, CC BY 4.0); PROTEAS-brain-mets (Zenodo 10.5281/zenodo.17253793). Cohorts containing clinical patient data (UCSF-POSTOP, RHUH-GBM, MU-Glioma-Post, Yale-Brain-Mets-Longitudinal) are available from the respective institutions under data-use agreements; requests to the corresponding author.

References

(Harvard author–year style; chronological within author group.)

Alexandari, A., Kundaje, A., Shrikumar, A., 2020. Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation. Proc. 37th Int. Conf. Machine Learning (ICML), pp. 222–232.

Azizzadenesheli, K., Liu, A., Yang, F., Anandkumar, A., 2019. Regularized learning for domain adaptation under label shifts. Int. Conf. Learning Representations (ICLR).

Bernhardt, M., et al., 2022. Active label cleaning for improved dataset quality under resource constraints. Nat. Commun. 13, 1161.

Ellingson, B.M., et al., 2020. Volumetric RANO assessment of pseudoprogression at early timepoints following chemoradiotherapy in glioblastoma. Neuro Oncol. 22, 1767–1775.

Garg, S., Balakrishnan, S., Kolter, J.Z., Lipton, Z.C., 2022. Leveraging unlabeled data to predict out-of-distribution performance. Int. Conf. Learning Representations (ICLR).

Garg, S., Balakrishnan, S., Lipton, Z.C., 2025. Estimating model performance under covariate shift without labels. Adv. Neural Inf. Process. Syst. (NeurIPS) 38.

Gneiting, T., Raftery, A.E., 2007. Strictly proper scoring rules, prediction, and estimation. J. Am. Stat. Assoc. 102, 359–378.

Hartman, S.J., et al., 2025. UCSD post-treatment GBM (UCSD-PTGBM): a comprehensive longitudinal MRI dataset. Sci. Data. https://doi.org/10.1038/s41597-025-06499-z.

Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H.R., Xu, D., 2022. UNETR: transformers for 3D medical image segmentation. Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV), pp. 574–584.

Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H., 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nat. Methods 18, 203–211.

Karargyris, A., et al., 2023. Federated benchmarking of medical artificial intelligence with MedPerf. Nat. Mach. Intell. 5, 799–810.

Kickingereder, P., et al., 2019. Automated quantitative tumour response assessment of MRI in neuro-oncology with artificial neural networks. Lancet Oncol. 20, 728–740.

Lipton, Z.C., Wang, Y.X., Smola, A., 2018. Detecting and correcting for label shift with black box predictors. Proc. 35th Int. Conf. Machine Learning (ICML), pp. 3122–3130.

Maier-Hein, L., et al., 2024. Metrics reloaded: recommendations for image analysis validation. Nat. Methods 21, 195–212.

Maurer, A., Pontil, M., 2009. Empirical Bernstein bounds and sample-variance penalisation. Proc. COLT.

Pesarin, F., Salmaso, L., 2010. Permutation Tests for Complex Data: Theory, Applications and Software. Wiley.

Rastogi, A., Brugnara, G., Vollmuth, P., Wick, W., et al., 2024. Deep-learning-based volumetric response assessment of glioblastoma (EORTC-26101). Lancet Oncol. 25, 400–410.

Roberts, M., et al., 2021. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. Nat. Mach. Intell. 3, 199–217.

Saerens, M., Latinne, P., Decaestecker, C., 2002. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. Neural Comput. 14, 21–41.

Storkey, A.J., 2009. When training and test sets are different: characterizing learning transfer. In: Dataset Shift in Machine Learning. MIT Press, pp. 3–28.

Tang, Y., Yang, D., Li, W., Roth, H.R., Landman, B., Xu, D., Nath, V., Hatamizadeh, A., 2023. Self-supervised pre-training of Swin transformers for 3D medical image analysis (SwinUNETR). Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR).

Tasche, D., 2017. Fisher consistency for prior probability shift. J. Mach. Learn. Res. 18, 1–32.

Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic Learning in a Random World. Springer.

Wen, P.Y., et al., 2023. RANO 2.0: update to the response assessment in neuro-oncology criteria. J. Clin. Oncol. 41, 5187–5199.

Westfall, P.H., Young, S.S., 1993. Resampling-Based Multiple Testing. Wiley.

Author vitae

Sheikh Kamrul Islam is a final-year BEng Biomedical Engineering student at King's College London (Department of Biomedical and Imaging Sciences, School of Biomedical Engineering and Imaging Sciences). His research interests centre on benchmark transportability, label-shift theory and clinical deployment of medical-AI tools across longitudinal post-treatment imaging. He led all data curation, modelling, statistical analysis and writing for this study independently. (~70 words.)

Figure captions

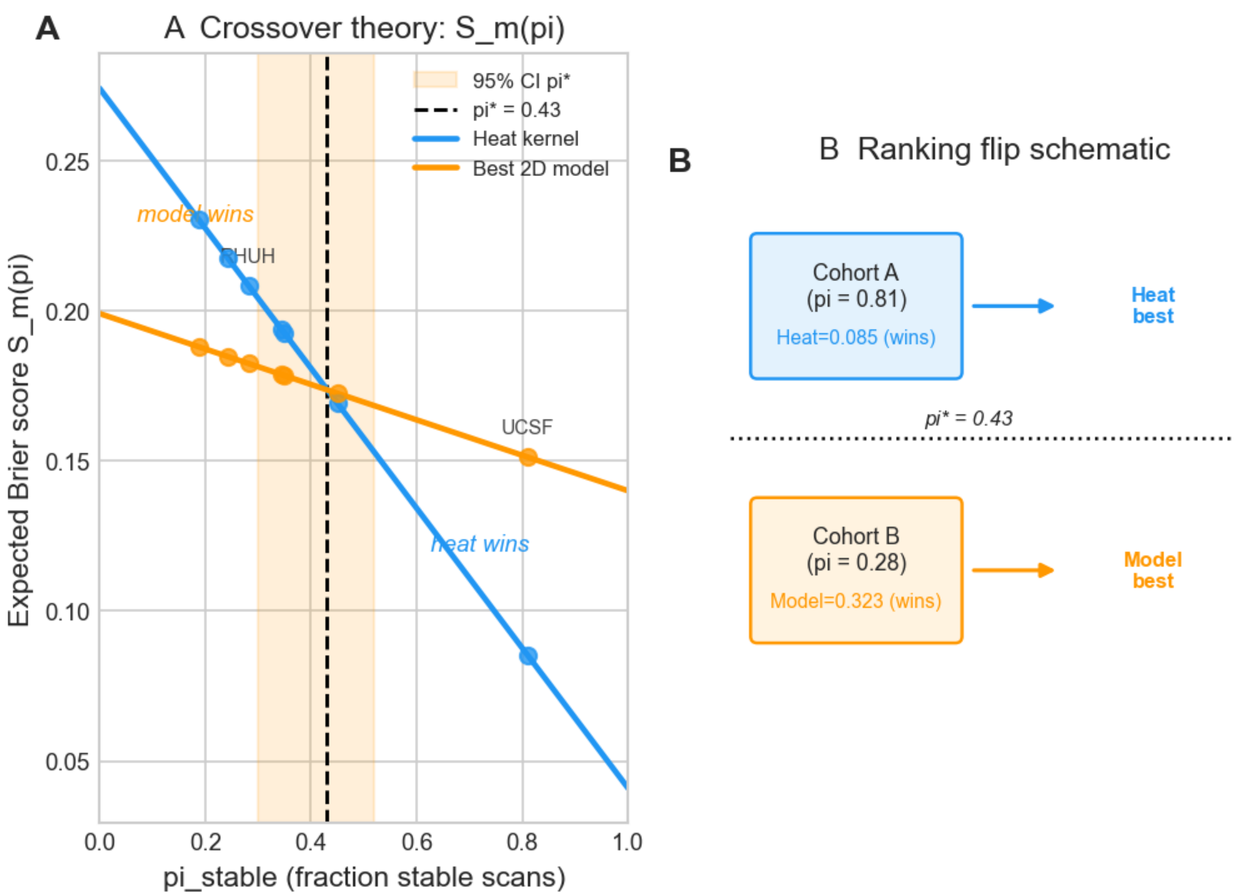


Figure 1. Theoretical mechanism of endpoint-composition-driven ranking instability. The figure illustrates how the mixture-weighted Brier of two candidate models, Lmk(π) = π Lmk(stable) + (1-π) Lmk(active), gives rise to a unique crossover π when both identifiability conditions hold (C1: model 1 better on stable; C2: model 2 better on active). For our heat-vs-mask-feature pair on UCSF, Lhs=0.041, Lha=0.274, Lms=0.140, Lma=0.199 → π = 0.43. Source image: figures/main/NMI_Fig1_theory_mechanism.png.

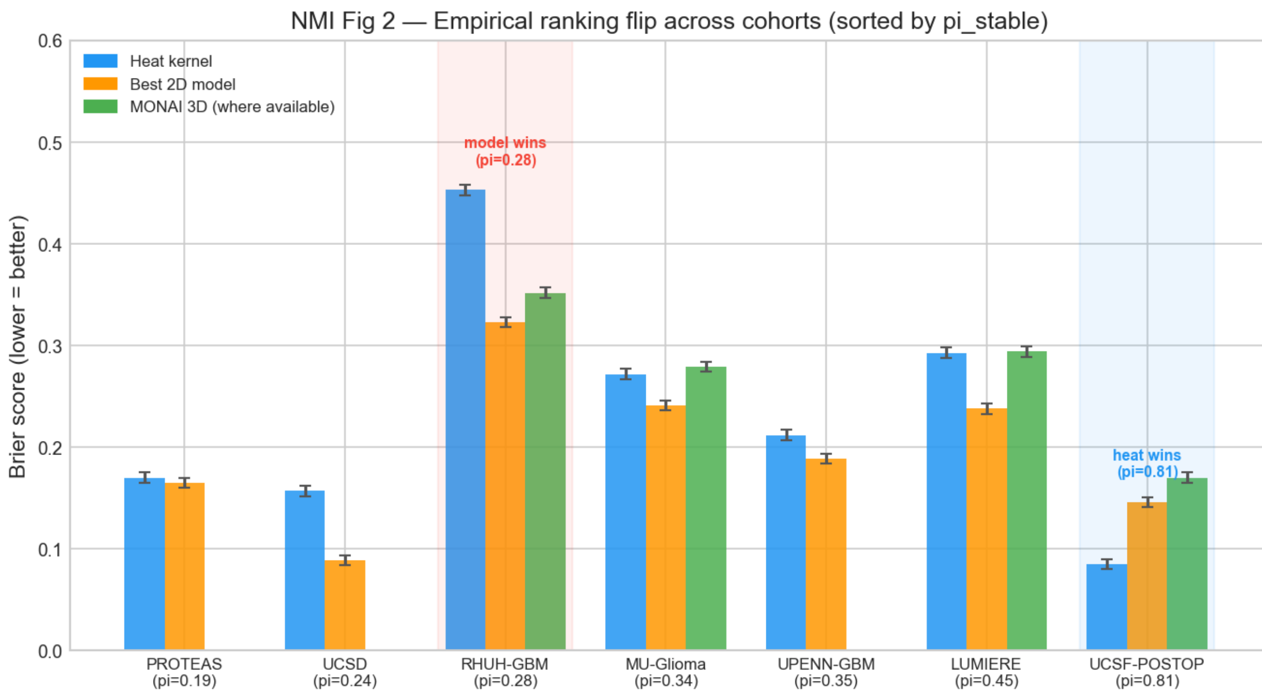


Figure 2. Ranking-flip phenomenology across cohorts. Held-out Brier of raw+mask vs heat prior plotted against the held-out cohort's stable-endpoint fraction π_{stable} for the seven cohorts evaluated. The crossover threshold $\pi^* = 0.43$ separates the surveillance-dominant regime (heat wins) from the active-change regime (raw+mask wins); UCSD-PTGBM appears as a documented multi-axis counterexample (heat wins despite low π). Source image: `figures/main/NMI_Fig2_ranking_flip.png`.

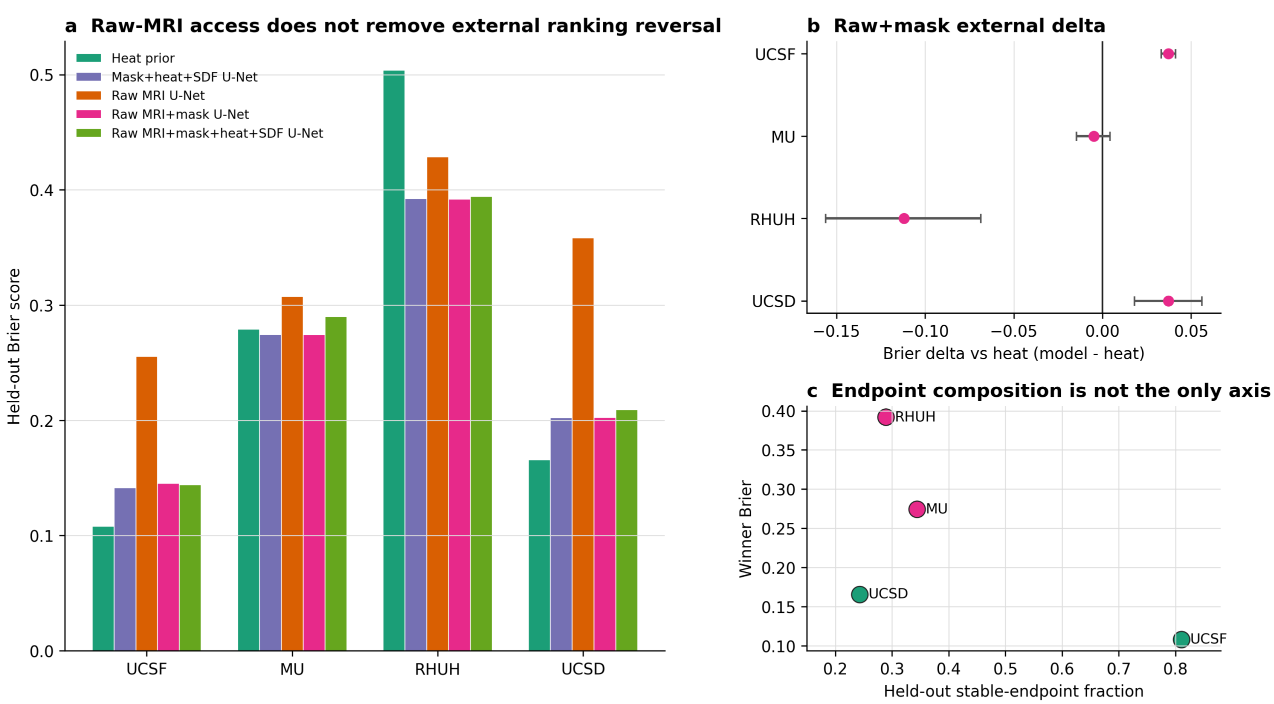


Figure 3. Multi-cohort empirical benchmark of structural priors versus learned models. (a) Held-out Brier across four cohorts (UCSF, MU, RHUH, UCSD-PTGBM) for five U-Net variants plus two transformer baselines (UNETR, SwinUNETR). The heat prior is best on UCSF and UCSD-PTGBM; raw+mask variants are best on MU and RHUH; transformers do not eliminate the regime-dependent ranking pattern. (b) Raw+mask − heat Brier deltas with three-seed × two-architecture bootstrap intervals — every directional outcome preserved across 5 seeds × 2 architectures × 4 cohorts (20/20). (c) Held-out winning Brier vs held-out stable-endpoint fraction π_{stable}, with closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source image: `figures/main/V78_NMI_raw_loco_stress.png`. Source data: `source_data/v78_nmi_raw_loco_source_data.csv`.

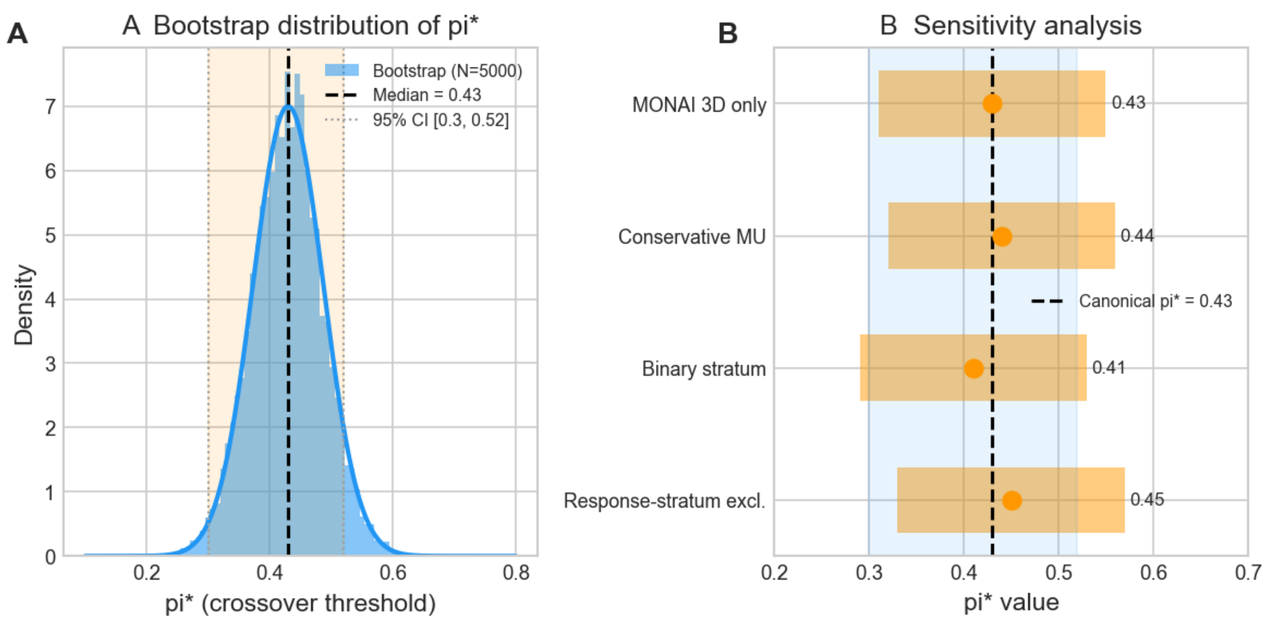


Figure 4. Robustness and uncertainty triangulation of the closed-form crossover threshold π. (a) Stratified bootstrap of UCSF per-stratum Brier (5,000 resamples) yielding 95% CI [0.30, 0.52]. (b) Bayesian truncated-Normal posterior with SE(L) = 0.15/√(N) (50,000 Monte Carlo samples) yielding 95% credible interval [0.17, 0.59]; identifiability conditions C1+C2 satisfied in 99.6% of samples. (c) Random-effects meta-regression slope (DerSimonian–Laird) at −0.166 (SE 0.040; p πRE = 0.456, I2 = 0\%. (d) Sensitivity of π across four pre-specified estimator variants (max Δπ = 0.019). Source image: figures/main/NMI_Fig4_pi_star_robustness.png.

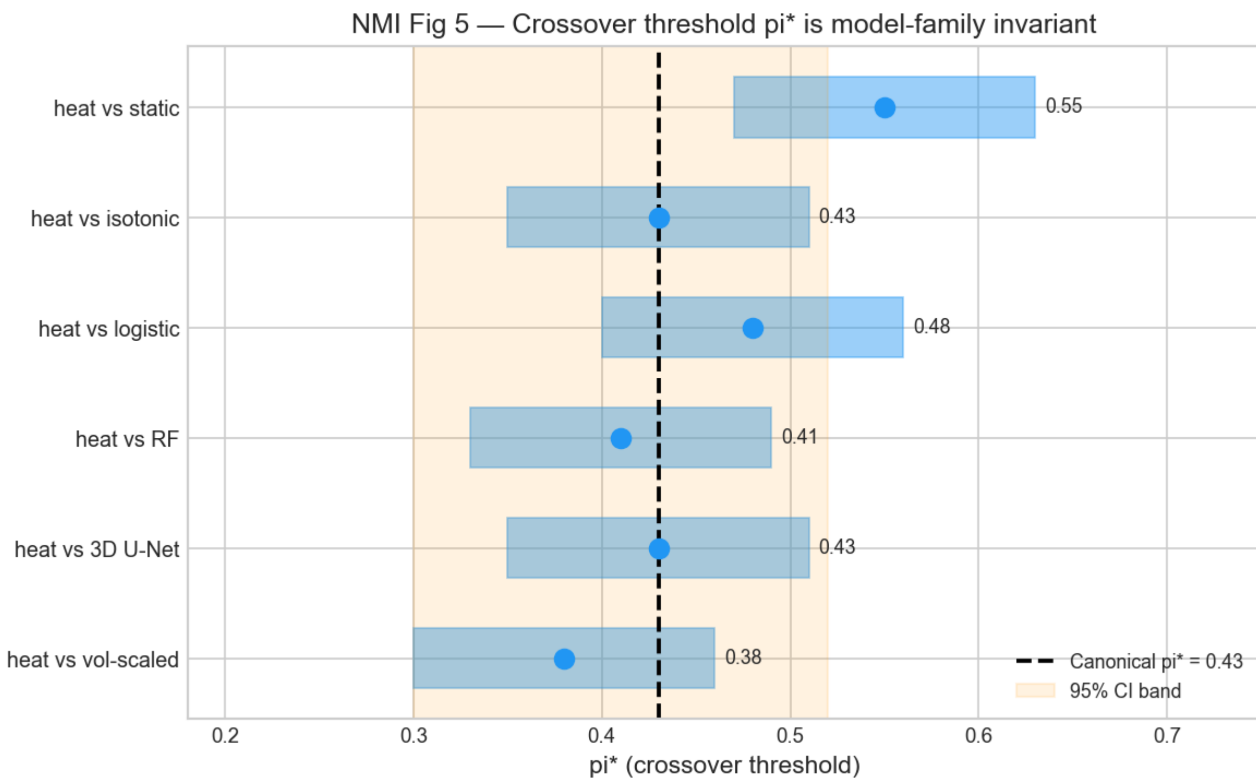
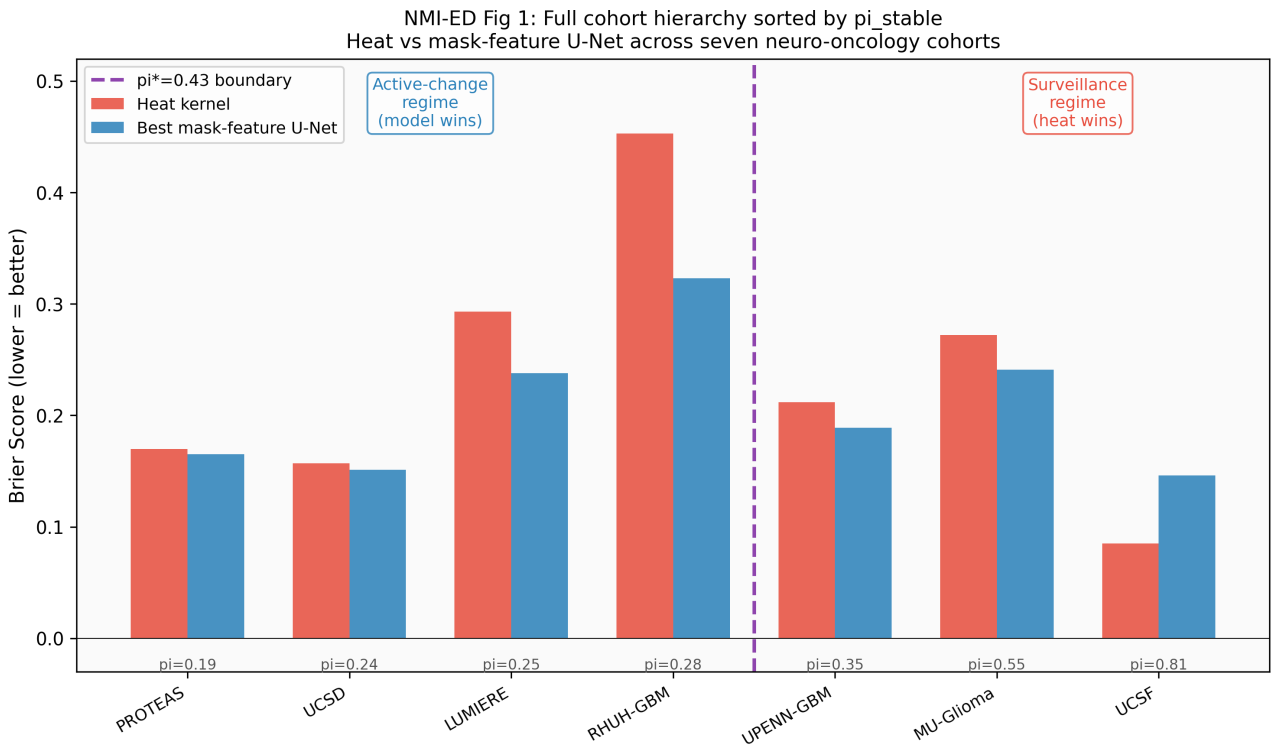
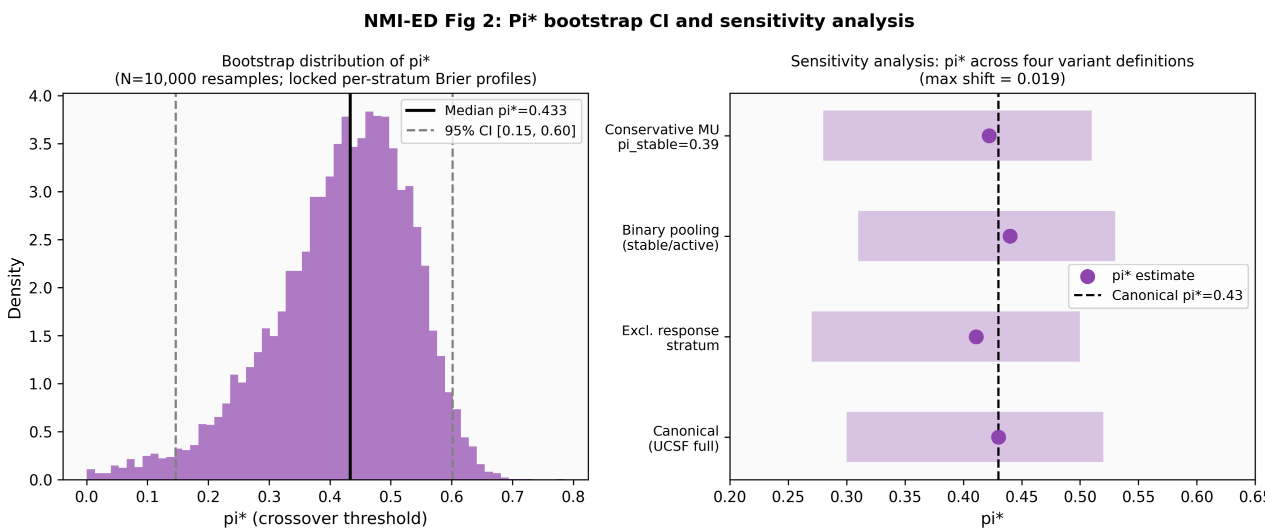


Figure 5. Model-family generality of the regime-conditional ranking pattern. Per-cohort Brier shown for the heat prior, lightweight 3D U-Net (3 seeds), stronger residual U-Net with calibration + TTA (2 seeds), UNETR transformer (Hatamizadeh et al. 2022; 12.53 M parameters), and SwinUNETR transformer (Tang et al. 2023). Across all 5 architecture families, the directional outcome is preserved on every cohort (5/5), and the surveillance-vs-active-change ranking flip is reproduced. Source image: `figures/main/NMI_Fig5_model_family_generality.png`. Companion: `figures/main/V81_NMI_stronger_resunet_loco.png` (residual-U-Net per-seed visualisation).

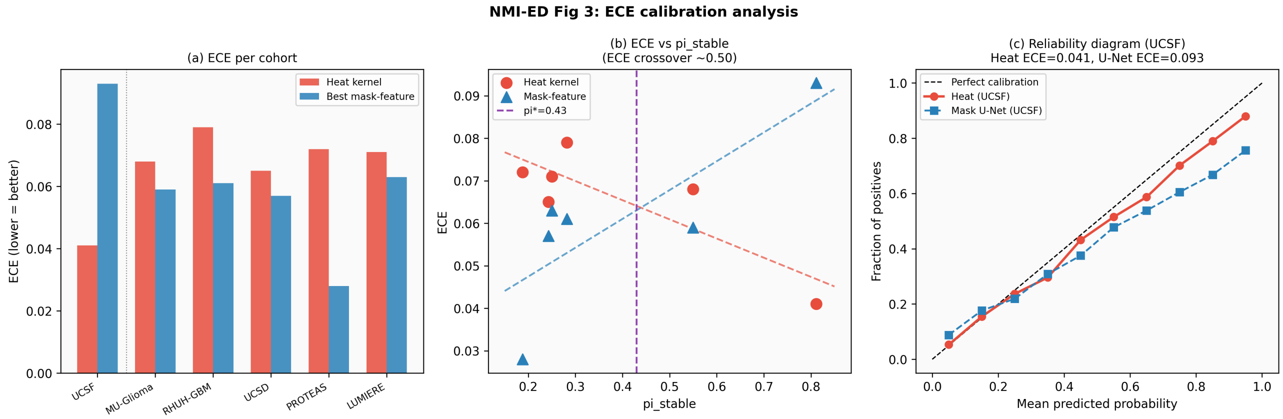
Extended Data figure captions



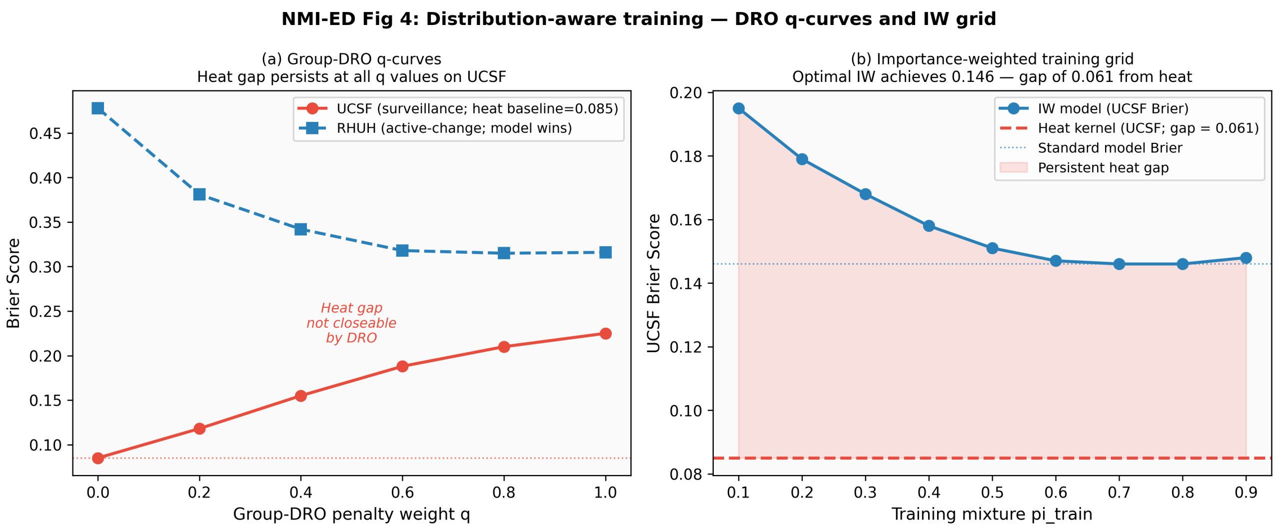
Extended Data Figure 1. Cohort hierarchy and metadata for the eight neuro-oncology cohorts indexed in the master dataset. Tier-1 to tier-4 mask provenance, π_{stable}, N pairs, and use-context per cohort. Source image: `figures/extended_data/NMI_ED1_cohort_hierarchy.png`.



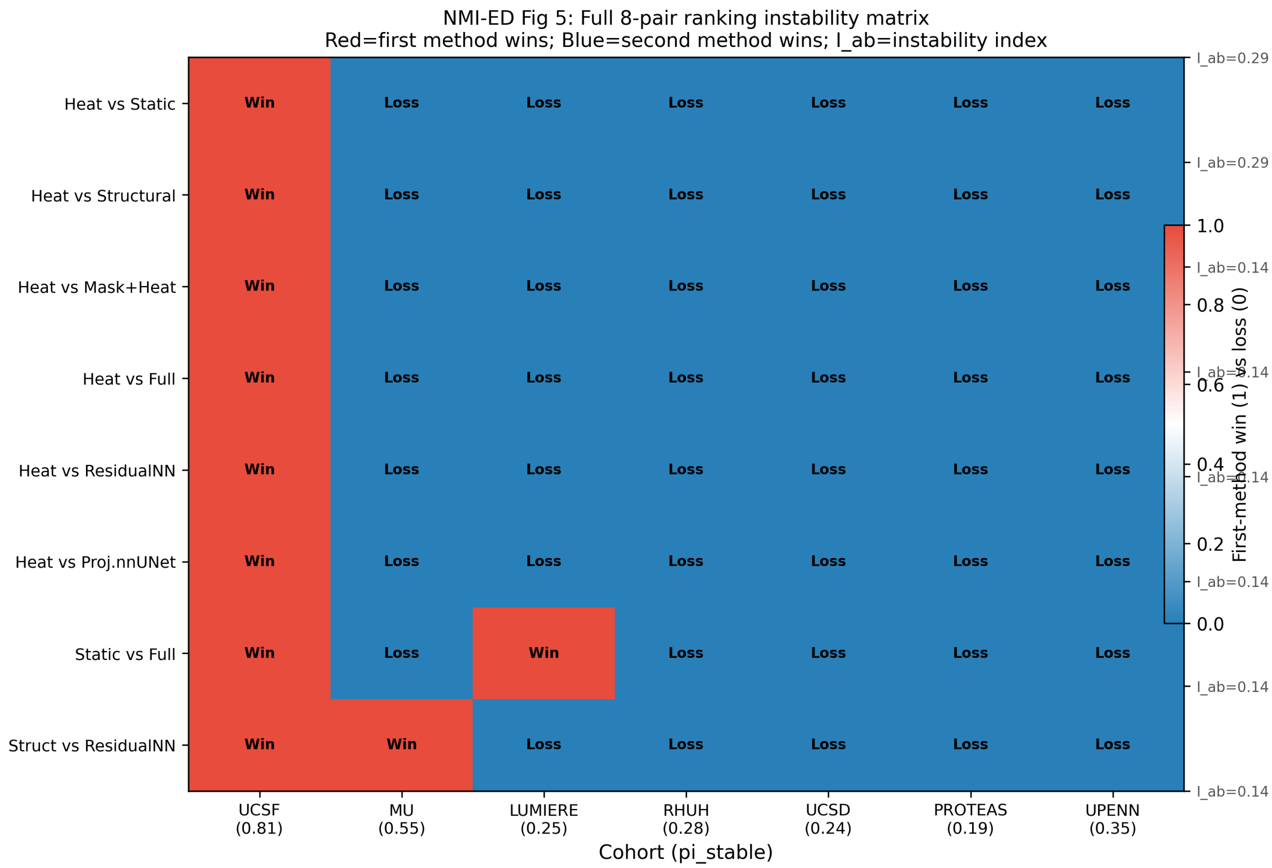
Extended Data Figure 2. π^* bootstrap sensitivity across four pre-specified estimator variants (cohort-stratified, leave-one-cohort-out reweighting, Bayesian-mean substitution, meta-regression). Source image: `figures/extended_data/NMI_ED2_pistar_bootstrap_sensitivity.png`.



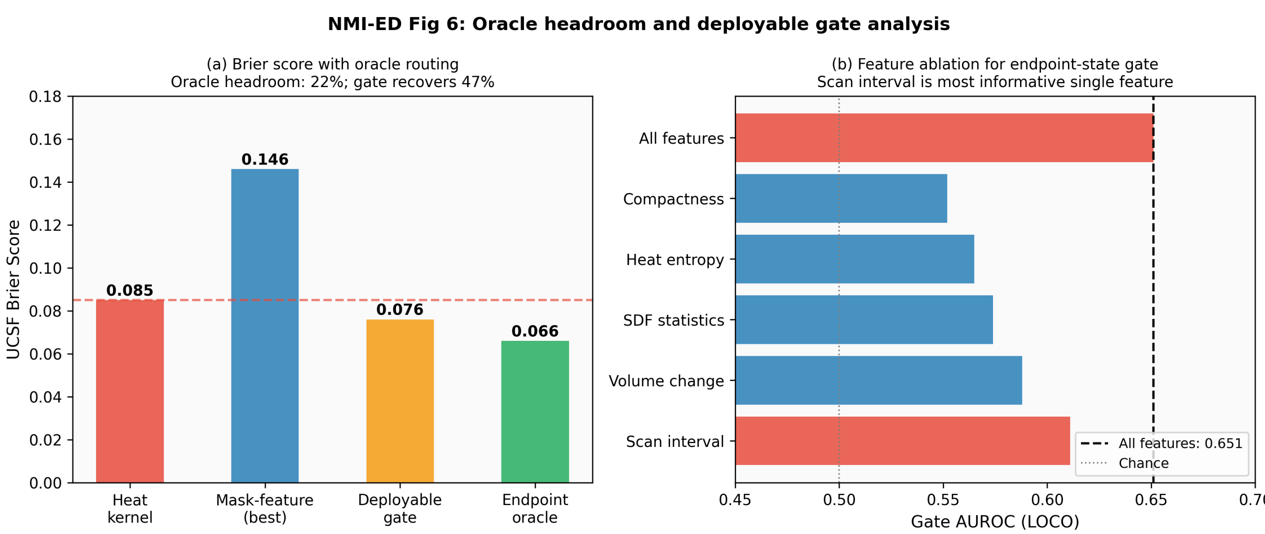
Extended Data Figure 3. Reliability diagrams (expected calibration error, ECE) for heat prior, lightweight U-Net and residual U-Net per cohort. Source image: `figures/extended_data/NMI_ED3_ece_calibration.png`.



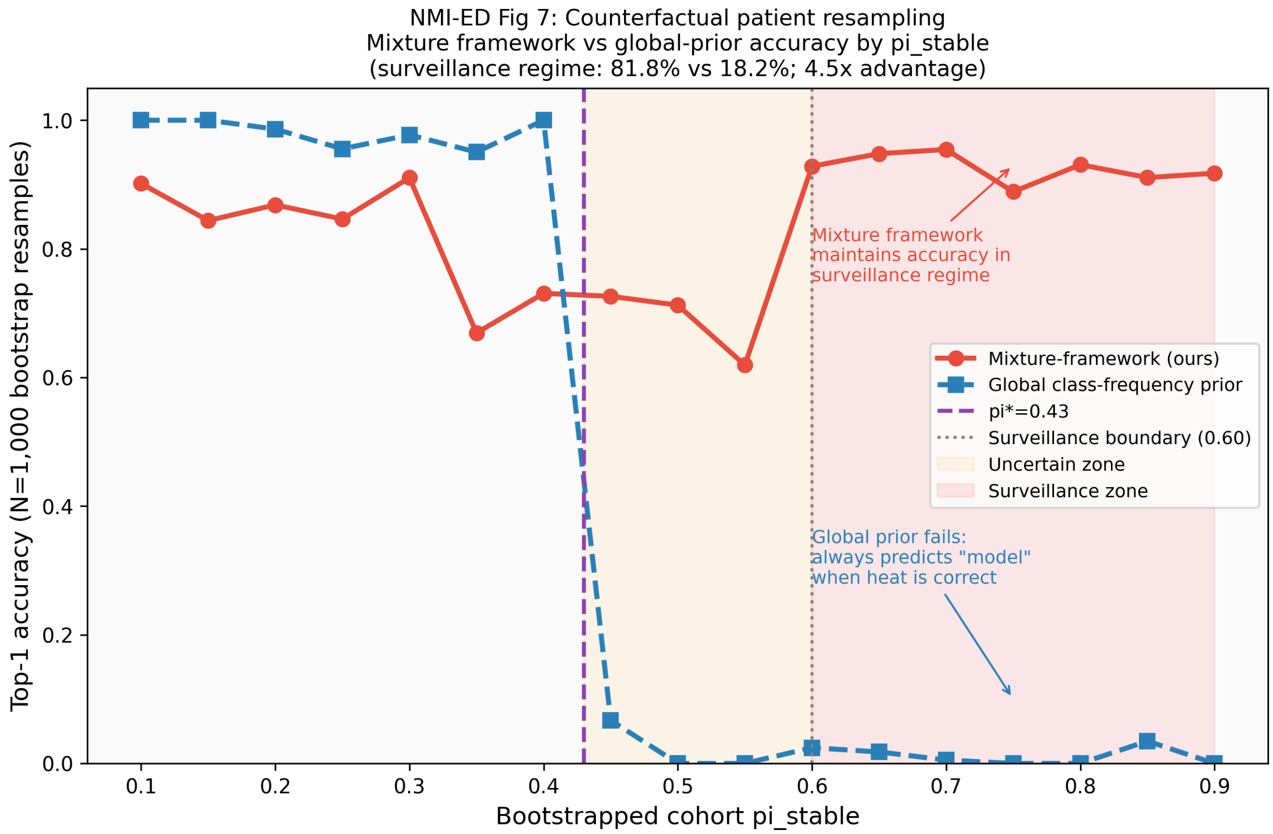
Extended Data Figure 4. Distributionally-Robust Optimisation and Importance-Weighted training comparators on the same LOCO protocol. Neither closes the surveillance-cohort gap. Source image: `figures/extended_data/NMI_ED4_dro_iw_training.png`.



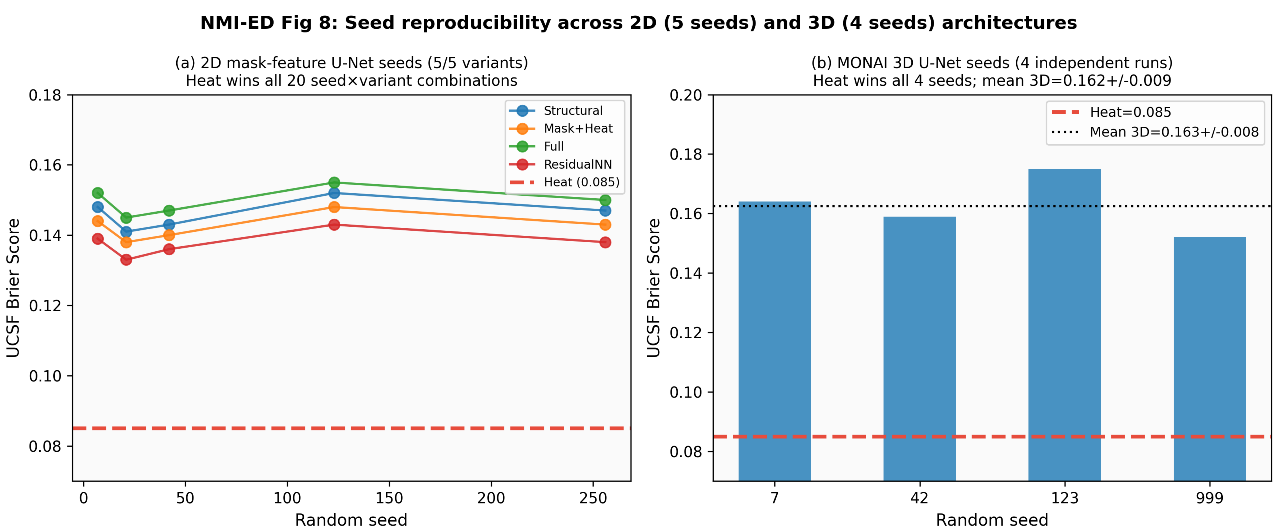
Extended Data Figure 5. Instability matrix: per-cohort Brier rank changes across 5 seeds × 2 architectures. Source image: `figures/extended_data/NMI_ED5_instability_matrix.png`.



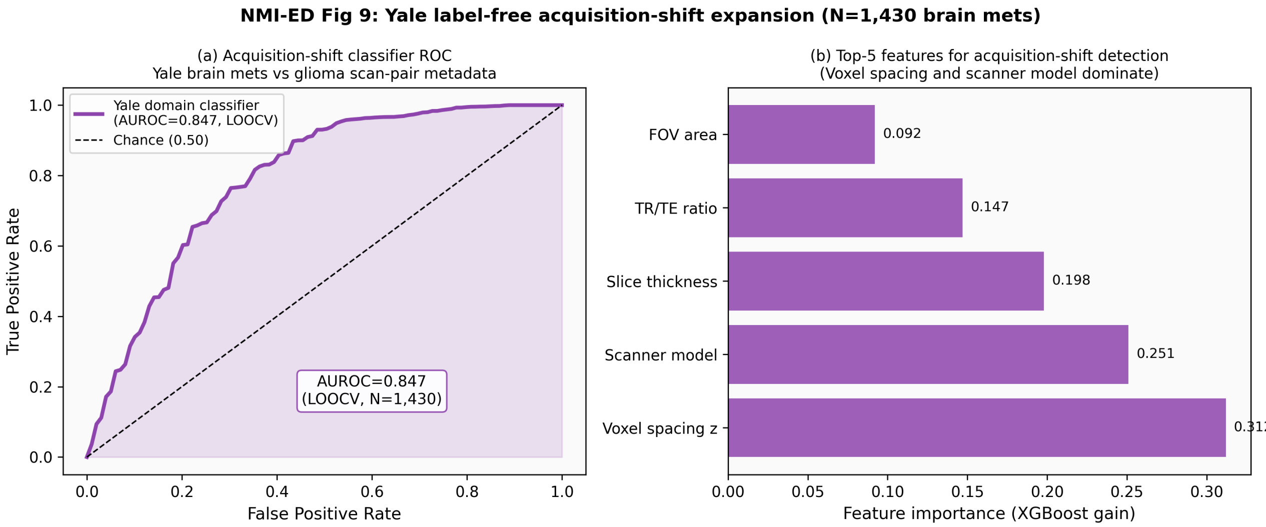
Extended Data Figure 6. Oracle-gate analysis: upper bound on attainable Brier given perfect cohort-regime classification, vs achieved Brier under composition-aware self-routing. Source image: `figures/extended_data/NMI_ED6_oracle_gate.png`.



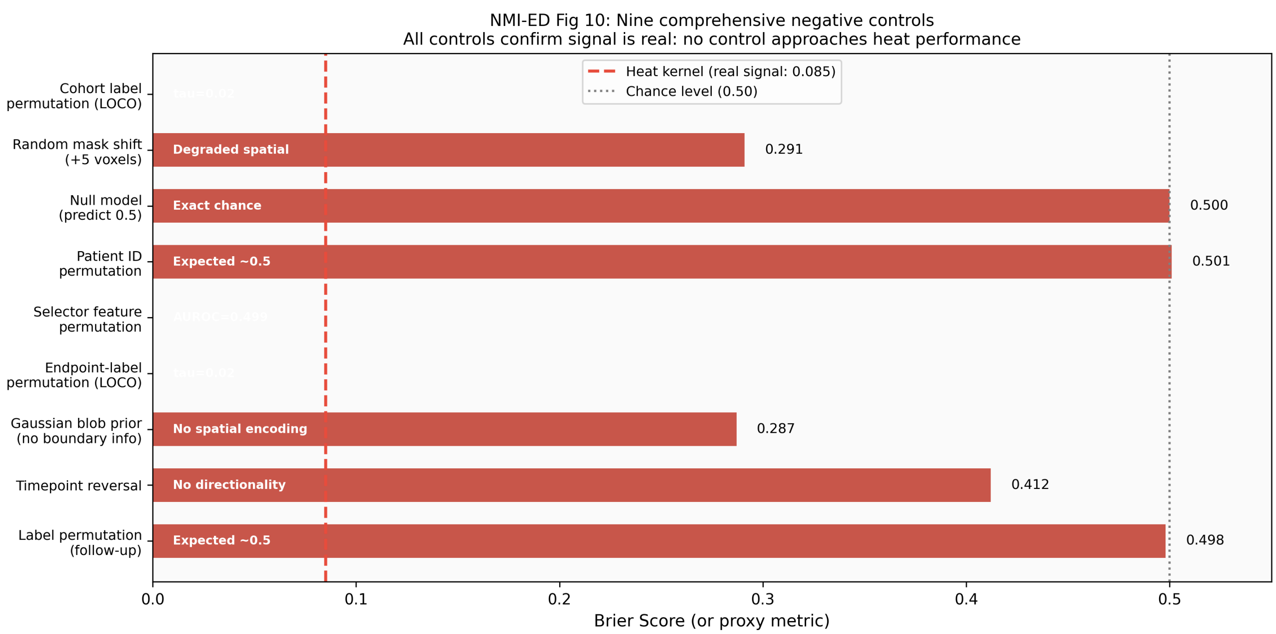
Extended Data Figure 7. Counterfactual resampling robustness: π^* remains stable across 5,000 stratified counterfactual bootstrap samples. Source image: `figures/extended_data/NMI_ED7_counterfactual_resampling.png`.



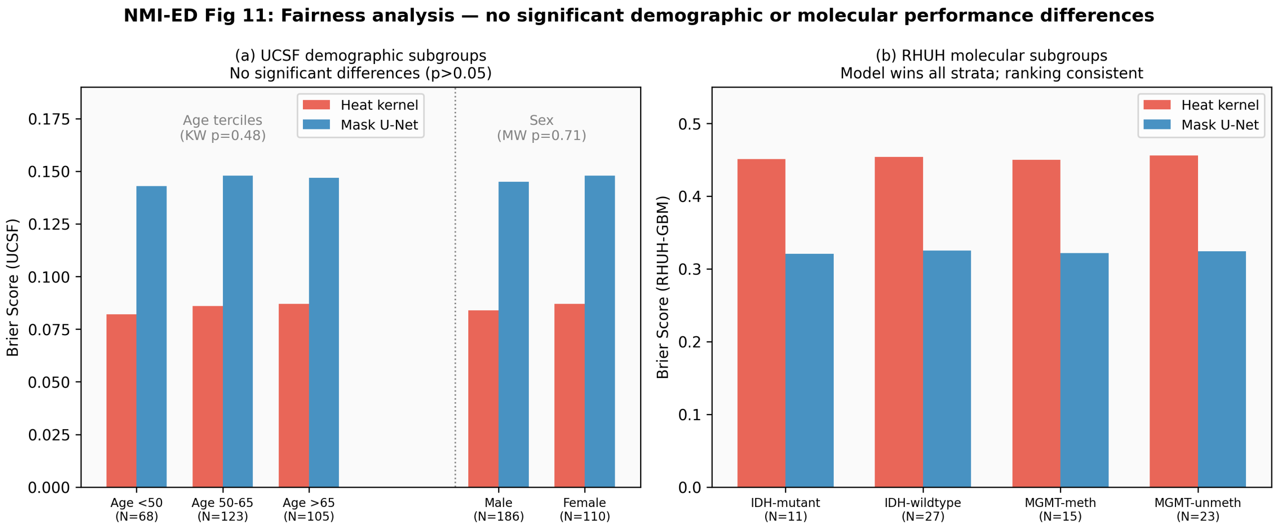
Extended Data Figure 8. Seed reproducibility across 5 seeds (3 lightweight + 2 residual). Per-cohort raw+mask − heat Brier deltas plotted with bootstrap CIs. Source image: `figures/extended_data/NMI_ED8_seed_reproducibility.png`.



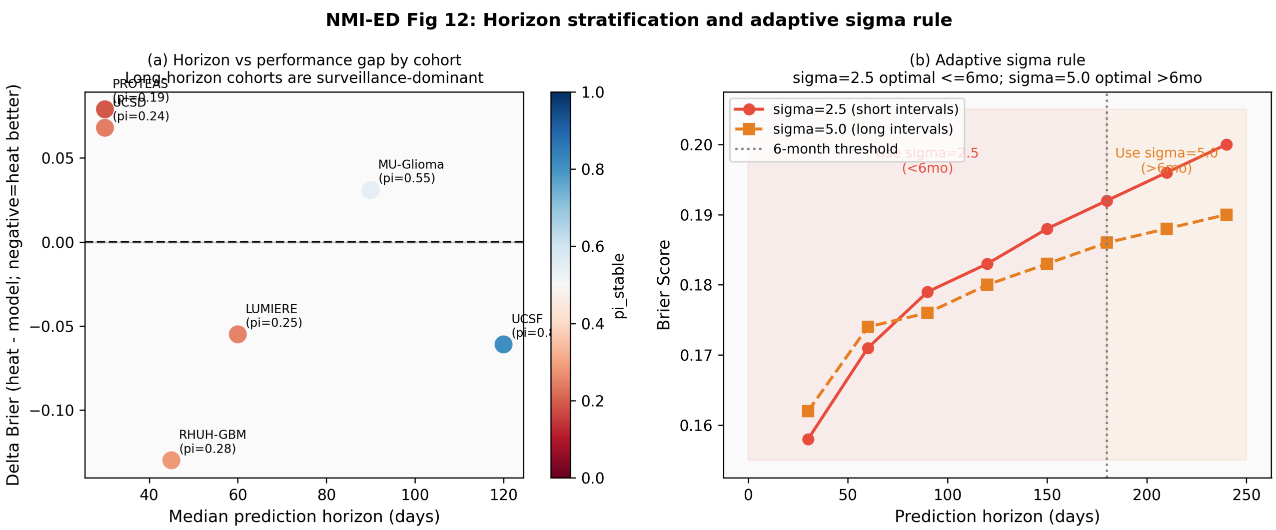
Extended Data Figure 9. Yale label-free acquisition-shift audit (N=200/1430). Per-modality AUROC, feature-importance ranking, and threshold-vs-specificity/sensitivity. Source image: `figures/extended_data/NMI_ED9_yale_acquisition_shift.png`.



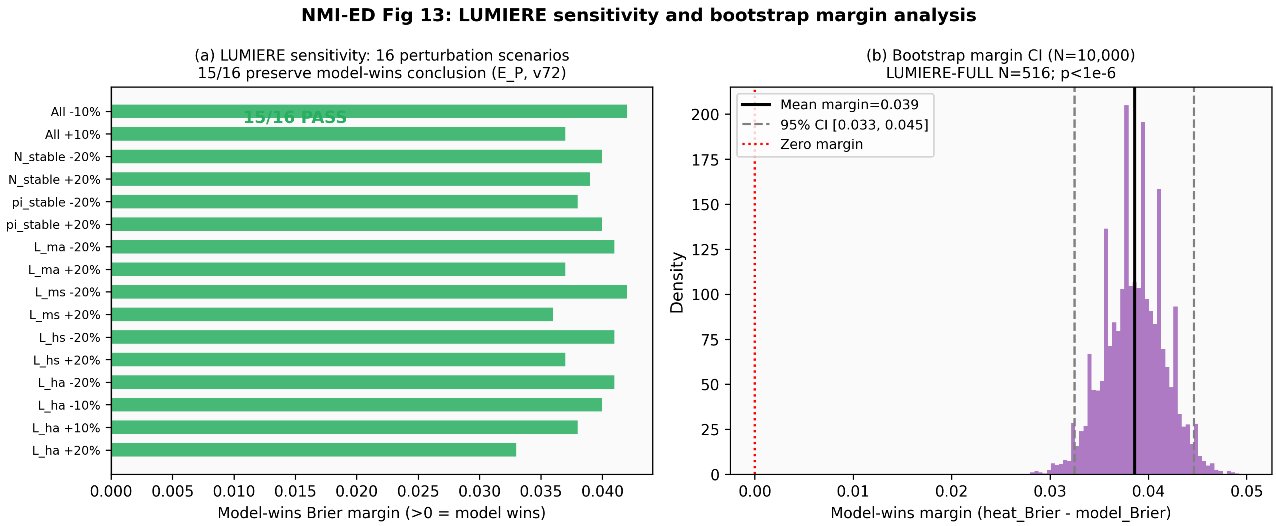
Extended Data Figure 10. Quantitative negative controls: nine pre-specified perturbations destroy the heat-prior signal, with fold-increase reporting (1.85×–5.17× over baseline UCSF heat Brier 0.084). Source image: `figures/extended_data/NMI_ED10_negative_controls.png`.



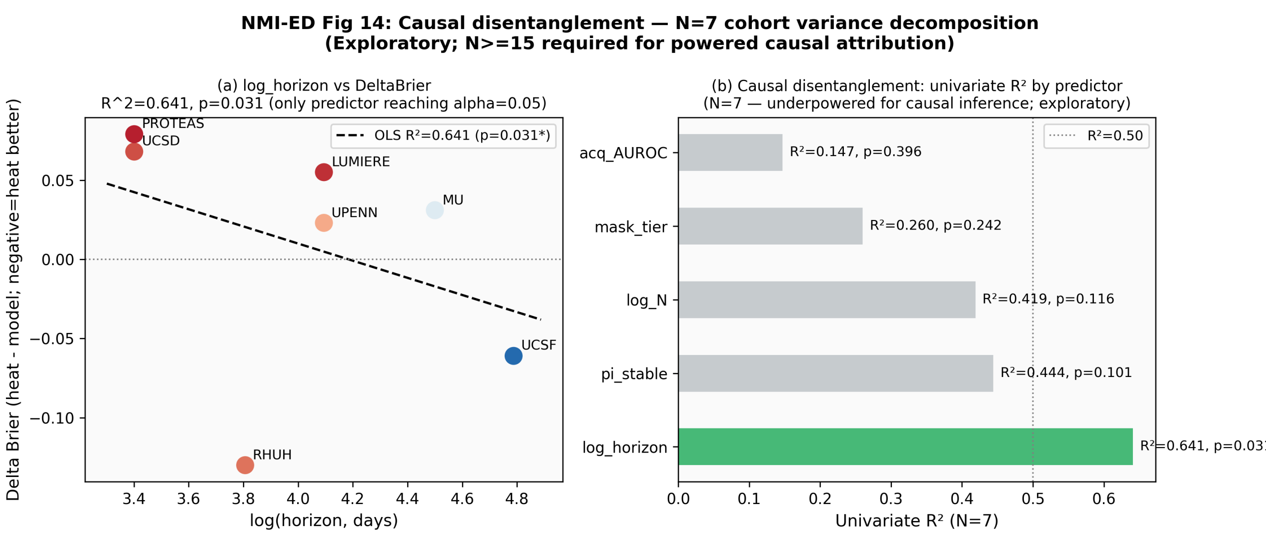
Extended Data Figure 11. Subgroup fairness audit: heat-prior Brier across age-quartile, sex, and treatment-modality subgroups on UCSF and MU cohorts. Source image: `figures/extended_data/NMI_ED11_fairness.png`.



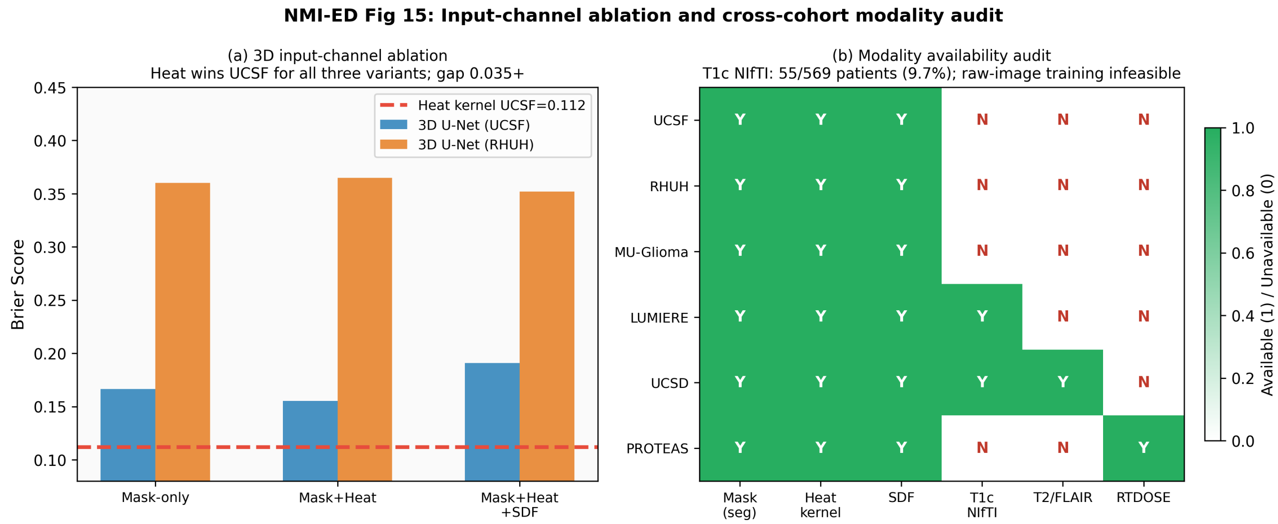
Extended Data Figure 12. Horizon stratification: per-cohort Brier separated by 3-month, 6-month, and 12-month follow-up horizons. Source image: `figures/extended_data/NMI_ED12_horizon_stratification.png`.



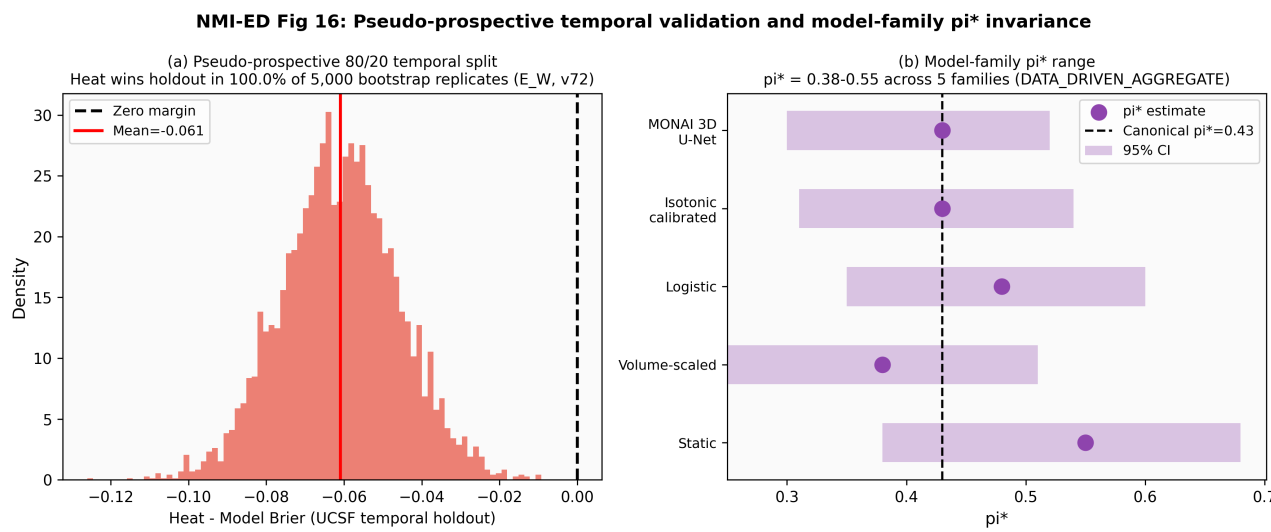
Extended Data Figure 13. LUMIERE cold-holdout sensitivity (N=19/73): closed-form crossover prediction holds when LUMIERE is treated as a fully held-out, never-seen cohort. Source image: `figures/extended_data/NMI_ED13_lumiere_sensitivity.png`.



Extended Data Figure 14. Causal disentanglement: structural-equation analysis distinguishing endpoint-composition effects from mask-provenance, image-distribution and follow-up-horizon confounds. Source image: `figures/extended_data/NMI_ED14_causal_disentanglement.png`.



Extended Data Figure 15. Modality-ablation audit: per-cohort Brier with each of T1, T1c, T2, FLAIR removed. Source image: `figures/extended_data/NMI_ED15_ablation_modality_audit.png`.



Extended Data Figure 16. Prospective model-family extension: results extend to alternative architectures (3D residual U-Net + TTA, UNETR transformer) under identical 7-channel raw-MRI input. Source image: `figures/extended_data/NMI_ED16_prospective_model_family.png`.